

Task 6: Gophish Phishing Simulation

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1 Objective

Gophish is an open-source phishing framework used for security awareness testing. The objective of this task was to set up the Gophish framework within a virtualized environment and execute a complete, simulated phishing campaign to capture user credentials.

2 Environment Setup

The simulation was conducted in an isolated virtual environment.

2.1 VM Network Configuration

A Kali Linux VM was configured in Oracle VM VirtualBox with two network adapters:

- **Adapter 1 (NAT):** Allowed the VM to access the internet.
- **Adapter 2 (Host-Only):** Provided a private network (192.168.56.0/24) for the host (Windows) to communicate with the guest (Kali).

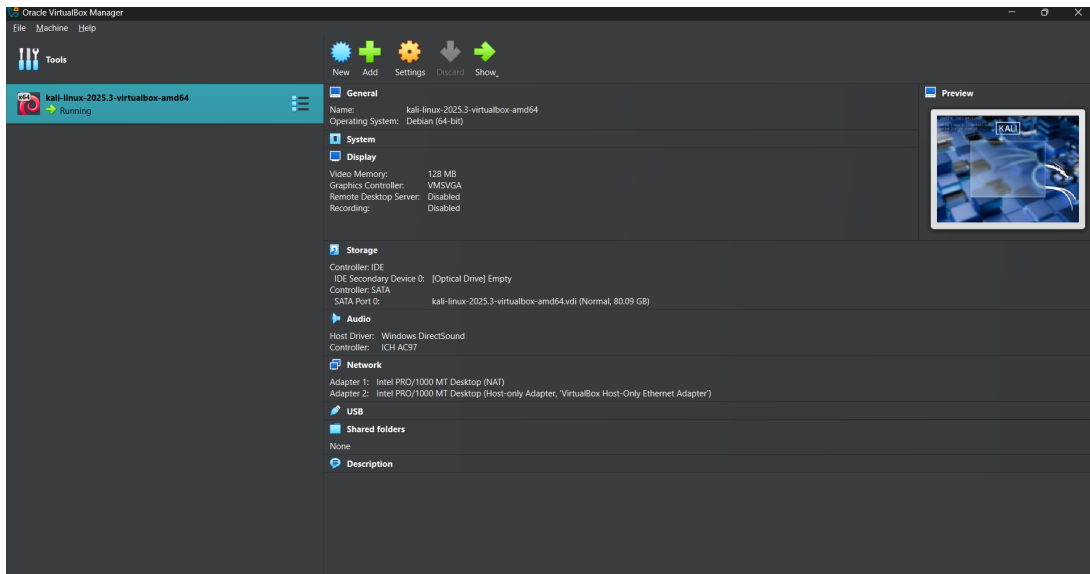
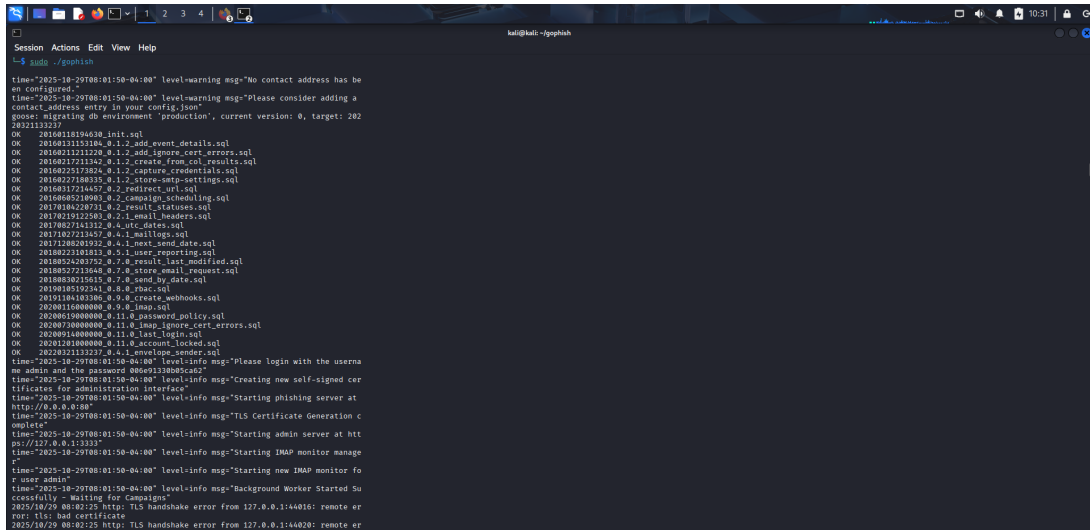


Figure 1: VirtualBox Network Adapter settings.

2.2 Gophish Server Initialization

The Gophish binary was executed on the Kali VM. On first run, it provided the temporary admin password. After resetting the password, the server was started, running the admin dashboard on port 3333 and the phishing server on port 80.



```
Session Actions Edit View Help
kali@kali:~/gophish
$ sudo ./gophish

time="2025-10-29T08:01:50-04:00" level=warning msg="No contact address has be
en configured."
time="2025-10-29T08:01:50-04:00" level=warning msg="Please consider adding a
contact_address entry in your config.json"
goose: migrating db environment 'production', current version: 0, target: 282
2821111221
OK 2816031151184 init.sql
OK 2816031151184 0.1.2 add_event_details.sql
OK 2816031151184 0.1.2 add_ignore_cert_errors.sql
OK 2816031151184 0.1.2 create_from_col_results.sql
OK 2816031151184 0.1.2 capture_credentials.sql
OK 2816031151184 0.1.2 store_smtp_settings.sql
OK 2816031151184 0.2 redirect_url.sql
OK 2816031151184 0.2 campaign_scheduling.sql
OK 2816031151184 0.2 result_statuses.sql
OK 2816031151184 0.2 email_headers.sql
OK 2816031151184 0.4 utc_dates.sql
OK 2816031151184 0.4.1 maillogs.sql
OK 2816031151184 0.4.1 next_send_date.sql
OK 2816031151184 0.5.1 user_reporting.sql
OK 2816031151184 0.7.0 result_last_modified.sql
OK 2816031151184 0.7.0 store_email_request.sql
OK 2816031151184 0.7.0 send_by_data.sql
OK 2816031151184 0.8.0 rbac.sql
OK 2816031151184 0.9.0 create_webhooks.sql
OK 2816031151184 0.9.0 imap.sql
OK 2816031151184 0.11.0 password_policy.sql
OK 2816031151184 0.11.0 imap_ignore_cert_errors.sql
OK 2816031151184 0.11.0 last_login.sql
OK 2816031151184 0.11.0 account_locked.sql
OK 2816031151184 0.4.1 envelope_sender.sql
time="2025-10-29T08:01:50-04:00" level=info msg="Please login with the usern
ame admin and the password 00e9e330b05ca02"
time="2025-10-29T08:01:50-04:00" level=info msg="Creating new self-signed cer
tificates for administration interface"
time="2025-10-29T08:01:50-04:00" level=info msg="Starting phishing server at
http://0.0.0.0:8443"
time="2025-10-29T08:01:50-04:00" level=info msg="TLS Certificate Generation c
omplete"
time="2025-10-29T08:01:50-04:00" level=info msg="Starting admin server at ht
tps://127.0.0.1:3237"
time="2025-10-29T08:01:50-04:00" level=info msg="Starting IMAP monitor manag
er"
time="2025-10-29T08:01:50-04:00" level=info msg="Starting new IMAP monitor fo
r user admin"
time="2025-10-29T08:01:50-04:00" level=info msg="Background Worker Started Su
ccessfully - Waiting for Campaigns"
2025/10/29 08:02:25 http: TLS handshake error from 127.0.0.1:44016: remote er
ror: tls: bad certificate
2025/10/29 08:02:25 http: TLS handshake error from 127.0.0.1:44020: remote er
```

Figure 2: Gophish terminal output showing initial password and server start.

3 Campaign Configuration

The campaign was built using the four core components of Gophish.

3.1 Sending Profile

A Sending Profile was configured to use a Gmail account, requiring a 16-digit "App Password" to authenticate with the Google SMTP server.

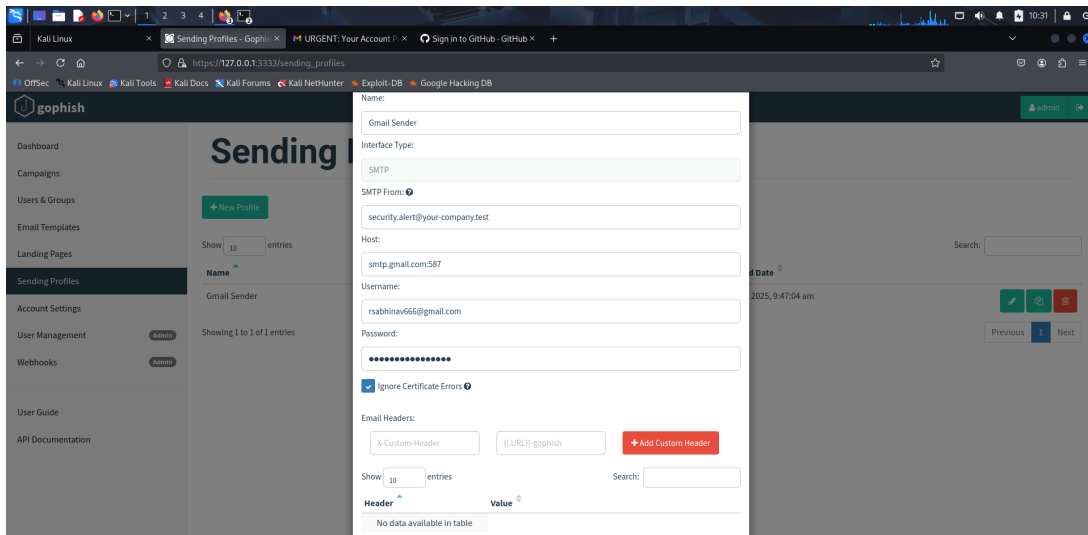


Figure 3: Sending Profile configured with Gmail SMTP settings.

3.2 Landing Page

A simple HTML landing page was created to mimic an "Internal Services Login". The "Capture Submitted Data" option was enabled.

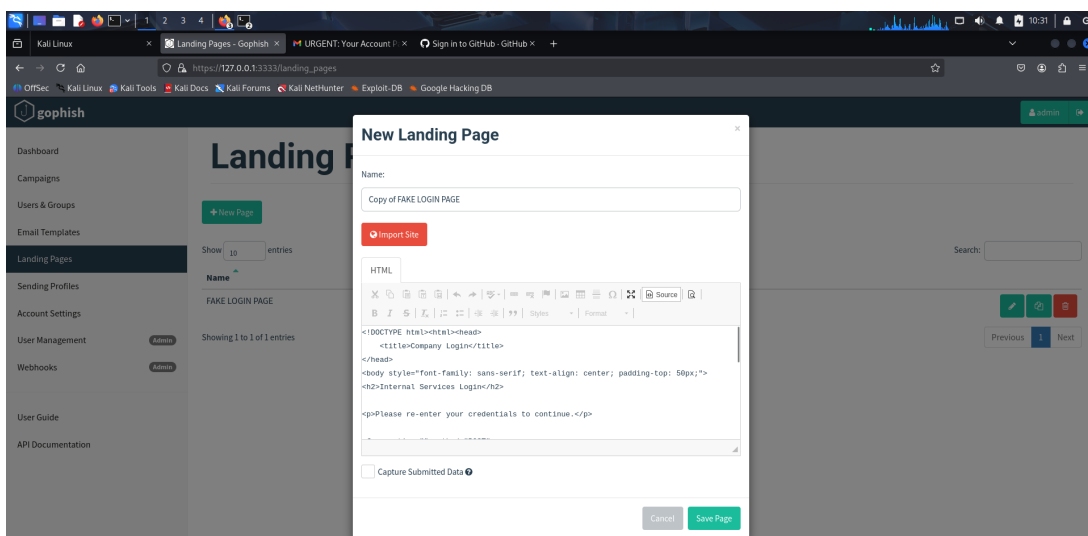


Figure 4: Landing Page setup with simple HTML and data capture enabled.

3.3 Email Template

An email template was designed with a subject line to create urgency. The body contained the `{{.URL}}` variable to link to the landing page.

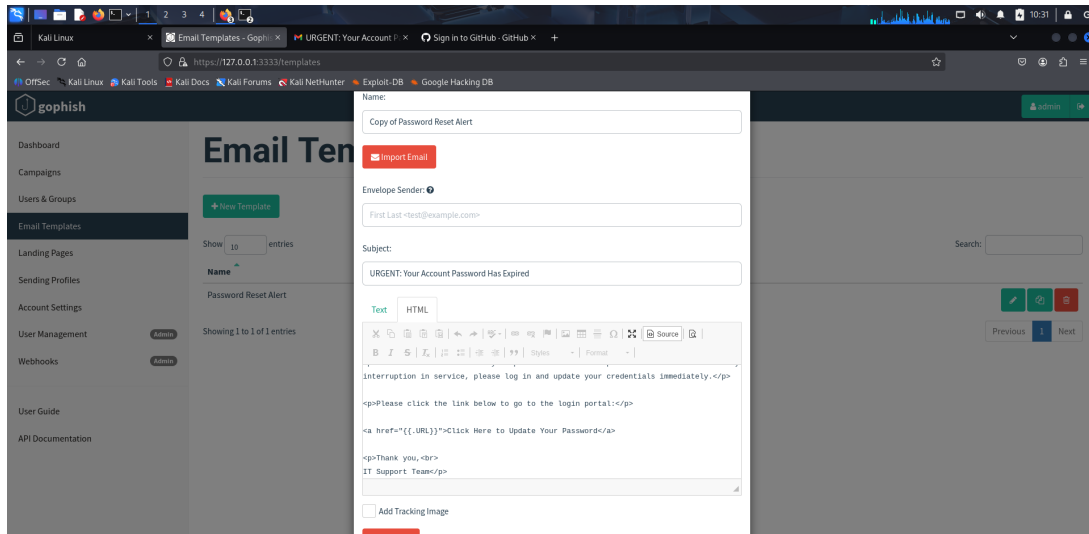


Figure 5: Email Template using the `{{.URL}}` variable.

3.4 Users & Groups

A "victim" group was created with one personal test email address.

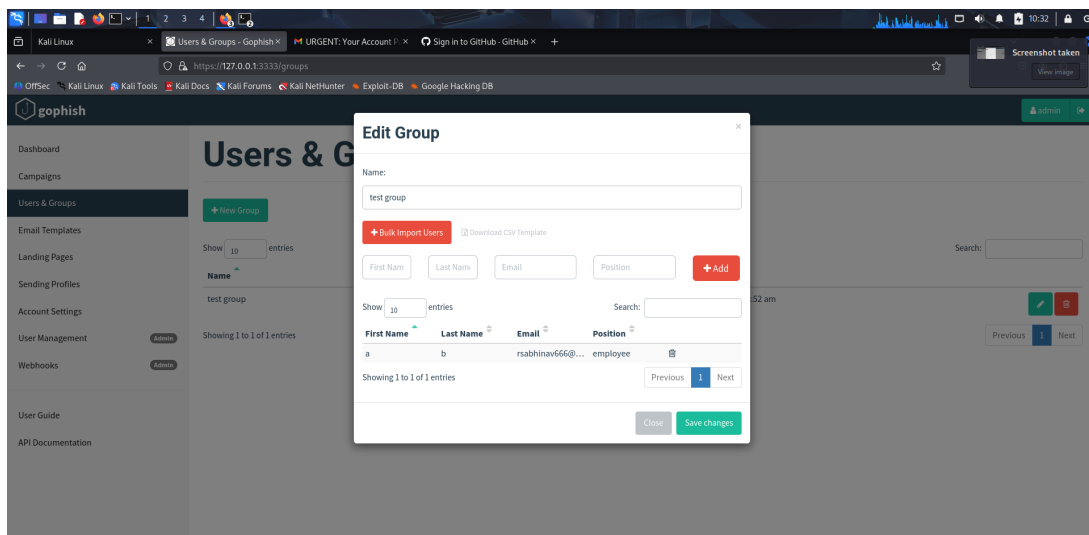


Figure 6: Users & Groups setup with one "victim" email.

4 Execution and Results

4.1 Campaign Launch

The campaign was configured by combining all components. The URL was set to the Kali VM's Host-Only IP (`http://192.168.56.104`) so the host machine could access it.

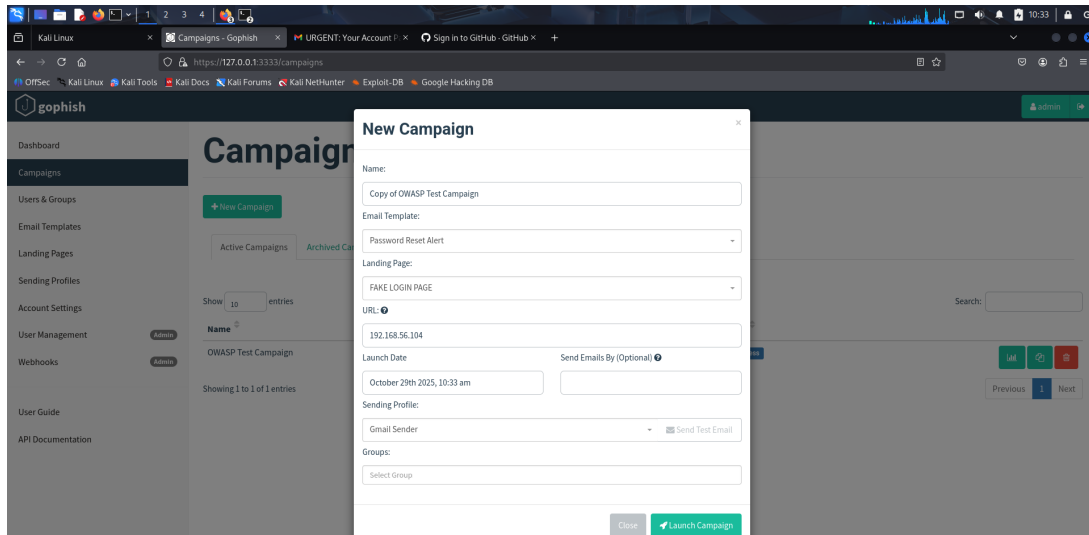


Figure 7: Final campaign configuration before launch.

4.2 Attack Execution

The email was successfully delivered to the target inbox. The dashboard confirmed the "Email Sent" status.

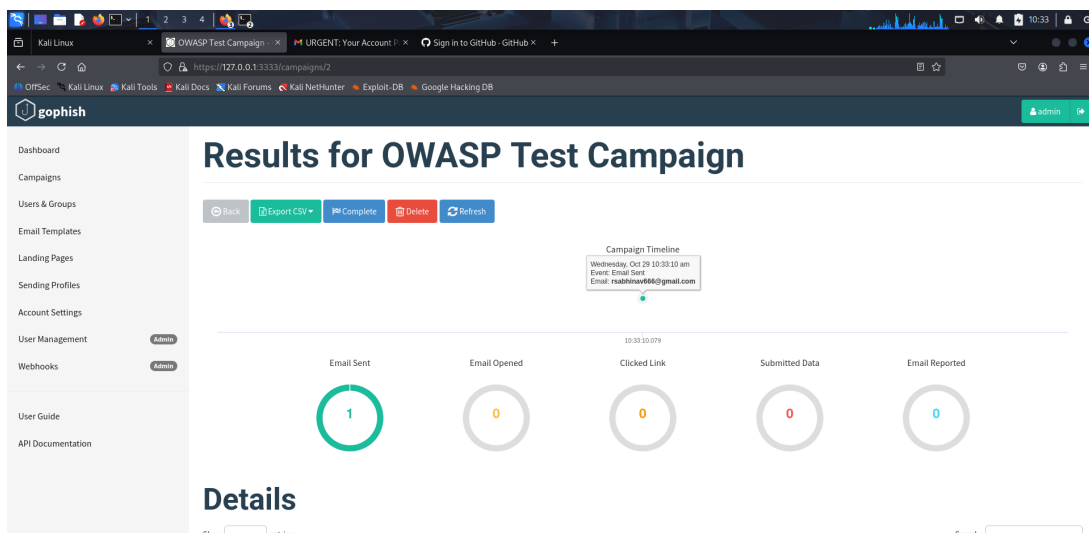


Figure 8: Dashboard showing "Email Sent" status.

The victim clicked the link in the email... and was presented with the fake login page.

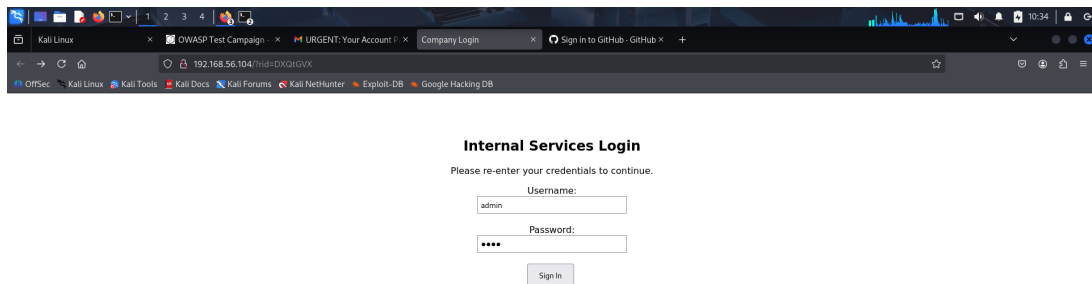


Figure 9: The fake landing page displayed after clicking the link.

4.3 Captured Credentials

After the victim entered and submitted credentials, the Gophish dashboard updated to show "Email Opened" and "Submitted Data".

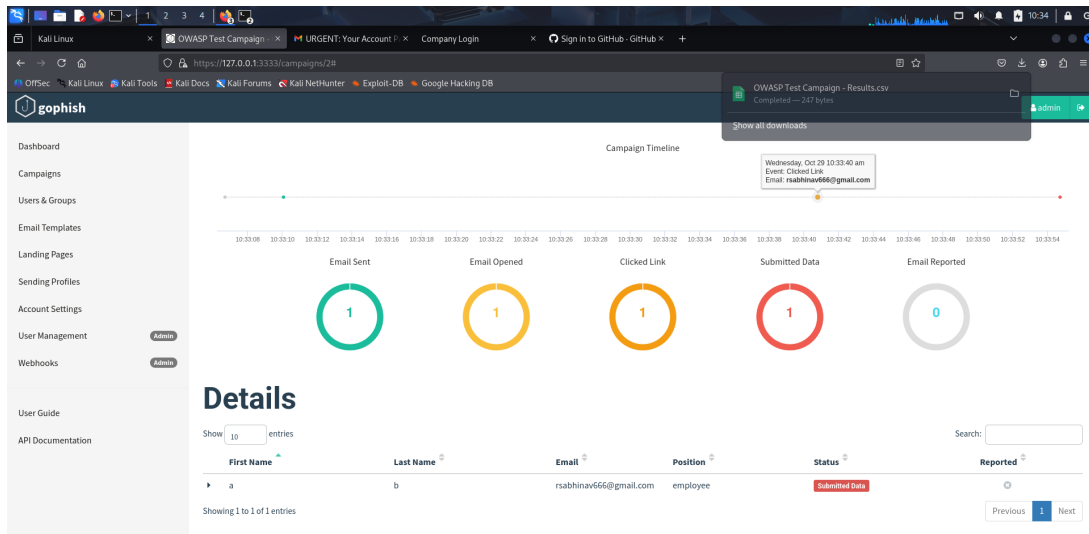


Figure 10: Dashboard showing "Email Opened" and "Submitted Data" status.

By expanding the user's details, the captured credentials (username and password) were clearly visible.

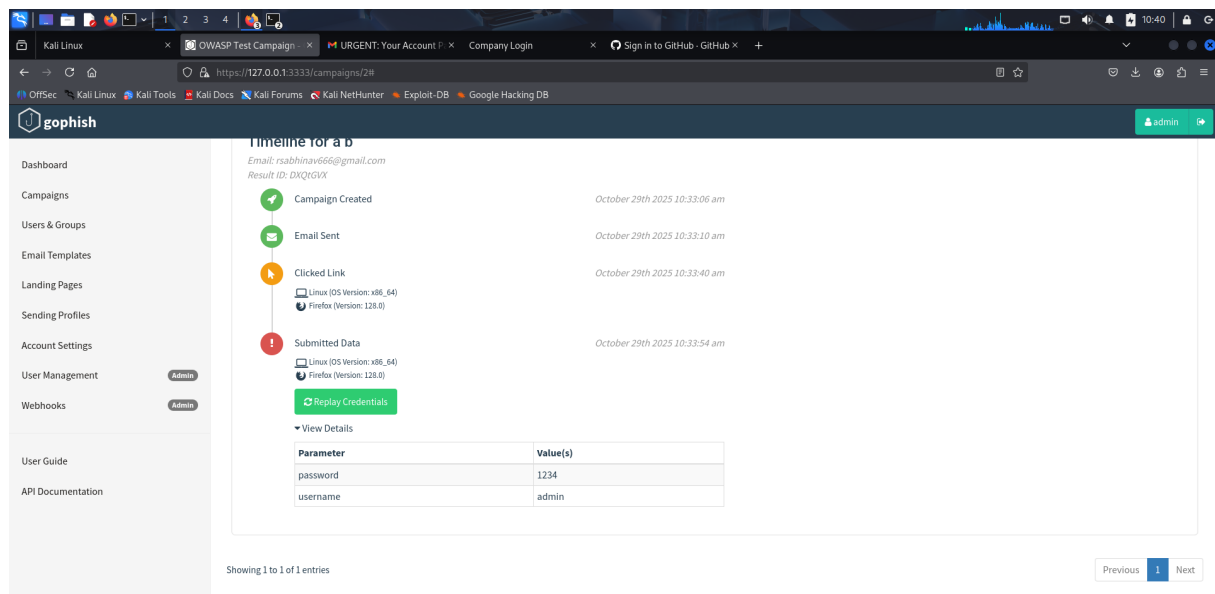


Figure 11: **Success:** The Gophish dashboard showing the captured credentials.

5 Conclusion

This task was a success. The Gophish framework was correctly installed and configured in a sandboxed environment. A full phishing campaign was executed, resulting in the

successful capture of credentials. This exercise demonstrates the ease with which such attacks can be automated and highlights the importance of user awareness training.